Chapter 3
Parity and divisibility

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you will learn

- Unfold evenness, oddness, divisibility, and congruence into witness statements.
- Construct new witnesses by algebraic calculation.
- Prove and use rules for modular arithmetic.
- Use residue cases to cover all integers.
- Explain how a Bézout identity justifies certain cancellation steps.

The recurring proof pattern

Unpack the hypothesis $\longrightarrow$ calculate $\longrightarrow$ identify a witness for the conclusion.

3.1 Definitions tell us what must be proved

For an integer n:

$$\begin{aligned} n \text{ is even} &\iff \exists k \in \mathbb{Z},\ n = 2k, \\ n \text{ is odd} &\iff \exists k \in \mathbb{Z},\ n = 2k + 1. \end{aligned}$$

- To prove evenness or oddness, produce an integer k.
- To use evenness or oddness, introduce the promised k and its equality.
- Different existential statements may have different witnesses, even if both are written with the letter k.

Lean: `dsimp [Int.Odd]` displays the textbook's definition.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.1.

Worked examples: zero and negative integers count too

Claim	Witness	Verification
0 is even	$k = 0$	$0 = 2 \cdot 0$
-8 is even	$k = -4$	$-8 = 2(-4)$
9 is odd	$k = 4$	$9 = 2 \cdot 4 + 1$
-7 is odd	$k = -4$	$-7 = 2(-4) + 1$

Why the domain matters

The witnesses for -8 and -7 are integers, not natural numbers. Parity is not restricted to positive integers.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.1; methods of Examples 3.1.1–3.1.2.

Worked example: turn one oddness witness into another

If $n = 2k + 1$, then

$$5n - 2 = 5(2k + 1) - 2 = 2(5k + 1) + 1.$$

Thus $5n - 2$ is odd, with witness $5k + 1$.

```
example {n : Int} (h : Int.Odd n) : Int.Odd (5 * n - 2) := by
  dsimp [Int.Odd] at *
  cases h with
  | intro k hk =>
      use 5 * k + 1
      calc
        5 * n - 2 = 5 * (2 * k + 1) - 2 := by rw [hk]
        _ = 2 * (5 * k + 1) + 1 := by ring
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.1; method of Example 3.1.3.

Worked example: every integer has one of two parities

Every integer is even or odd. Use this fact to prove $n^2 + n$ is even.

Case 1: $n = 2k$

$$n^2 + n = (2k)^2 + 2k = 2(2k^2 + k).$$

Case 2: $n = 2k + 1$

$$n^2 + n = (2k + 1)^2 + (2k + 1) = 2(2k^2 + 3k + 1).$$

Both witnesses are integers, and the cases exhaust all integers. Lean supplies the case split as `Int.even_or_odd n`.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 3.1.9; parity dichotomy revisited in Chapter 4.

Practice 3.1: name the output witness

1. If n is even, prove that $n^2 + 3$ is odd.
2. If m, n are odd integers, prove that mn is odd.

Use different letters for witnesses from independent hypotheses. End each proof by explicitly identifying an integer k in the form $2k + 1$.

Practice 3.1: solution discussion

1. Write $n = 2r$ for some $r \in \mathbb{Z}$. Then

$$n^2 + 3 = 4r^2 + 3 = 2(2r^2 + 1) + 1.$$

The output witness is $2r^2 + 1$.

2. Write $m = 2a + 1$ and $n = 2b + 1$, with $a, b \in \mathbb{Z}$. Then

$$mn = (2a + 1)(2b + 1) = 2(2ab + a + b) + 1.$$

The output witness is $2ab + a + b$.

Source: Heather Macbeth, *The Mechanics of Proof*, Problem 2 is a standard parity fact also practiced in Section 3.1.

3.2 Divisibility is an existence statement

For $a, b \in \mathbb{Z}$, define

$$a \mid b \quad \iff \quad \exists k \in \mathbb{Z},\ b = ak.$$

We say that a divides b, that a is a factor of b, or that b is a multiple of a.

- $6 \mid 42$, because $42 = 6 \cdot 7$.
- $-3 \mid 12$, because $12 = (-3)(-4)$.
- $a \mid 0$ for every a, because $0 = a \cdot 0$.
- $0 \mid b$ forces $b = 0$.

For divisibility in $\mathbb{N}$, the witness must lie in $\mathbb{N}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.2.1–3.2.2 and Section 3.2 exercises.

Worked example: retain a common factor

If $a \mid b$, write $b = ak$. Then

$$b^2 + 2b = (ak)^2 + 2ak = a(k(ak + 2)).$$

This proves $a \mid b^2 + 2b$.

```
example {a b : Int} (h : Dvd.dvd a b) :
    Dvd.dvd a (b ^ 2 + 2 * b) := by
  cases h with
  | intro k hk =>
      use k * (a * k + 2)
      calc
        b ^ 2 + 2 * b = (a * k) ^ 2 + 2 * (a * k) := by rw [hk]
        _ = a * (k * (a * k + 2)) := by ring
```

Source: Heather Macbeth, *The Mechanics of Proof*, Example 3.2.3; ASCII form of the divisibility notation.

Two basic facts and the hypotheses they need

Transitivity. For integers, $a \mid b$ and $b \mid c$ imply $a \mid c$. Indeed, if $b = ar$ and $c = bs$, then

$$c = (ar)s = a(rs).$$

A size bound in $\mathbb{N}$

If $a, b \in \mathbb{N}$, $b > 0$, and $a \mid b$, then $0 < a \leq b$.

Write $b = ak$. Since $ak > 0$, both a and k are positive. Thus $k \geq 1$, and $a = a \cdot 1 \leq ak = b$.

The condition $b > 0$ is essential: $8 \mid 0$, but $8 \nleq 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.2; witness composition and Examples 3.2.7–3.2.8.

Worked example: prove non-divisibility by a gap

Consecutive multiples test

Let $d > 0$ and $q, t \in \mathbb{Z}$. If $dq < t < d(q+1)$, then $d \nmid t$.

Why? If $t = dk$ for an integer k, dividing by positive d gives $q < k < q+1$, impossible for an integer.

Example. To show $4 \nmid -6$, observe

$$4(-2) = -8 < -6 < -4 = 4(-2+1).$$

The same test works for negative t; choose the two surrounding multiples.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.2; method of Example 3.2.6.

Practice 3.2: compose the witnesses

1. For $a, b \in \mathbb{Z}$, prove that $a \mid b$ implies $a \mid 3b - b^2$.
2. For $m, n, p \in \mathbb{Z}$, prove that $m \mid n$ and $n^3 \mid p$ imply $m^3 \mid p$.
3. Show that $5 \nmid 17$ by placing 17 between consecutive multiples.

Do not divide by a or m: those numbers may be zero. Use the equalities supplied by divisibility instead.

Practice 3.2: solution discussion

1. If $b = ak$, then

$$3b - b^2 = 3ak - a^2k^2 = a(3k - ak^2).$$

2. Write $n = mr$ and $p = n^3s$. Then

$$p = (mr)^3s = m^3(r^3s).$$

3. Since $5 \cdot 3 = 15 < 17 < 20 = 5(3 + 1)$, the number 17 is not a multiple of 5.

3.3 Congruence compares a difference to the modulus

For integers a, b, m, define

$$a \equiv b \pmod{m} \quad \Longleftrightarrow \quad m \mid (a - b).$$

Equivalently, $a - b = mk$ for some $k \in \mathbb{Z}$.

Worked example

$-8 \equiv 2 \pmod{5}$, because $-8 - 2 = 5(-2)$.

- Congruent numbers need not be equal.
- For $m > 0$, congruence means having the same remainder modulo m.
- The definition also makes sense for $m = 0$: it reduces to $a = b$.

Lean's textbook notation is represented by `Int.ModEq m a b`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.3 definition and Examples 3.3.1–3.3.2.

Addition and subtraction preserve congruence

Assume $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Choose integers r, s with $a - b = mr$ and $c - d = ms$. Then

$$\begin{aligned}(a + c) - (b + d) &= (a - b) + (c - d) \\ &= mr + ms = m(r + s),\end{aligned}$$

$$\begin{aligned}(a - c) - (b - d) &= (a - b) - (c - d) \\ &= mr - ms = m(r - s).\end{aligned}$$

Therefore $a + c \equiv b + d \pmod{m}$ and $a - c \equiv b - d \pmod{m}$. Negation is similar: $(-a) - (-b) = -(a - b) = m(-r)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.3.3–3.3.5.

Multiplication preserves congruence

Again assume $a - b = mr$ and $c - d = ms$. The useful identity is

$$ac - bd = (a - b)c + b(c - d).$$

It gives

$$\begin{aligned} ac - bd &= (mr)c + b(ms) \\ &= m(rc + bs). \end{aligned}$$

Hence $ac \equiv bd \pmod{m}$, with witness $rc + bs$.

A useful special case

$a^2 - b^2 = (a - b)(a + b) = m(r(a + b))$, so $a^2 \equiv b^2 \pmod{m}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.3.6 and 3.3.8.

Congruence supports chains and powers

For integers a, b, c, m:

- **Reflexivity:** $a - a = m \cdot 0$, so $a \equiv a \pmod{m}$.
- **Symmetry:** if $a - b = mr$, then $b - a = m(-r)$.
- **Transitivity:** if $a - b = mr$ and $b - c = ms$, then $a - c = m(r + s)$.

Power rule

If $a \equiv b \pmod{m}$ and $k \in \mathbb{N}$, then $a^k \equiv b^k \pmod{m}$.

For $k = 0$ both sides are 1. Repeated multiplication proves each positive-power case; Chapter 6 organizes this argument as induction on k.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.3.8–3.3.10, Section 3.3 exercises; Example 6.1.3.

Ordinary division is not a congruence rule

Modulo 6 we have

$$2 \cdot 1 \equiv 2 \cdot 4 \pmod 6,$$

because $2 - 8 = -6$ is divisible by 6. But cancelling the 2 would assert

$$1 \equiv 4 \pmod 6,$$

which is false: $1 - 4 = -3$ is not divisible by 6.

What failed?

An equality can be divided by a nonzero real factor. A congruence needs additional arithmetic information about the factor and modulus.

Section 3.5 supplies identities that make certain cancellations valid.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.3; new counterexample illustrating Example 3.3.7.

Practice 3.3: prove the rule from the definition

Suppose $a \equiv b \pmod{m}$.

1. Prove $2a + 1 \equiv 2b + 1 \pmod{m}$ by giving a witness.
2. Prove $a^3 \equiv b^3 \pmod{m}$ using a factorization.

Start from $a - b = mr$. Your calculation should end with m times an integer.

Practice 3.3: solution discussion

Choose $r \in \mathbb{Z}$ with $a - b = mr$. **1.**

$$(2a + 1) - (2b + 1) = 2(a - b) = 2mr = m(2r).$$

2.

$$\begin{aligned} a^3 - b^3 &= (a - b)(a^2 + ab + b^2) \\ &= m(r(a^2 + ab + b^2)). \end{aligned}$$

The witnesses are $2r$ and $r(a^2 + ab + b^2)$, respectively.

Source: Heather Macbeth, *The Mechanics of Proof*, Problem 2 follows Example 3.3.9.

3.4 Calculate with the rules you have proved

Let $a, b \in \mathbb{Z}$, with $a \equiv 3 \pmod 7$ and $b \equiv 4 \pmod 7$. Find the residue of $a^2 + b$.

$$\begin{aligned} a^2 + b &\equiv 3^2 + 4 \pmod 7 && \text{power and addition rules} \\ &= 13 && \text{arithmetic} \\ &\equiv 6 \pmod 7 && 13 - 6 = 7 \cdot 1. \end{aligned}$$

Why this is a proof

Each congruence uses an established rule; an equality also gives a congruence. Transitivity connects the initial expression to its final residue.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.4; method of Example 3.4.2.

Worked example: cover every possible residue

Claim. Every integer x satisfies $x^3 \equiv x \pmod 3$.
There are three possible residues of x modulo 3:

Residue r	r^3	$r^3 - r$	Divisible by 3?
0	0	0	$0 = 3 \cdot 0$
1	1	0	$0 = 3 \cdot 0$
2	8	6	$6 = 3 \cdot 2$

In each case $x^3 \equiv r^3 \equiv r \equiv x \pmod 3$. The table is exhaustive because every integer has one of these residues.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 3.4.4.

Lean: congruence transport and numerical verification

```
-- In the textbook's modular-arithmetic environment:
example {a b : Int}
    (ha : Int.ModEq 7 a 3) (hb : Int.ModEq 7 b 4) :
    Int.ModEq 7 (a ^ 2 + b) (3 ^ 2 + 4) := by
  rel [ha, hb]

example : Int.ModEq 7 13 6 := by
  use 1
  numbers
```

The chapter file also imports `Library.Tactic.ModEq`.
For residue cases, `mod_cases hx : x % 3` creates the three cases used on the preceding slide.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.4 and its companion Lean file.

Practice 3.4: substitution, witnesses, and cases

1. If $n \equiv 2 \pmod 5$, show $n^2 + 3n + 1 \equiv 1 \pmod 5$.
2. Find an integer k such that $4k \equiv 3 \pmod 7$.
3. Show that every integer square is congruent to 0 or 1 modulo 4.

Use a different method in each problem: substitute, choose a witness, and split into residues.

Practice 3.4: solution discussion

1. Substitute the known residue:

$$n^2 + 3n + 1 \equiv 2^2 + 3 \cdot 2 + 1 = 11 \equiv 1 \pmod 5.$$

2. Take $k = 6$, since $4 \cdot 6 - 3 = 21 = 7 \cdot 3$.

3. For residues $r = 0, 1, 2, 3$ modulo 4:

$$r^2 = 0, 1, 4, 9 \quad \text{has residue} \quad 0, 1, 0, 1.$$

The power rule transfers this calculation from r back to any integer having residue r.

3.5 Expressing 1 as an integer combination

The integers 3 and 7 satisfy

$$(-2) \cdot 3 + 1 \cdot 7 = 1.$$

Multiplying by any integer n gives

$$n = -2(3n) + 7n.$$

Why this helps

If $3n$ is a multiple of 7, then both terms on the right are multiples of 7. Their sum n must therefore also be a multiple of 7.

A relation expressing 1 as an integer combination of two numbers is a Bézout identity for those numbers.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.5; method of Example 3.5.1.

Worked example: justify a cancellation by an explicit witness

Assume $n \in \mathbb{Z}$ and $7 \mid 3n$. Choose $t \in \mathbb{Z}$ with $3n = 7t$. Then

$$\begin{aligned} n &= -2(3n) + 7n && (-2) \cdot 3 + 7 = 1 \\ &= -2(7t) + 7n && \text{the divisibility witness} \\ &= 7(n - 2t) && \text{algebra.} \end{aligned}$$

Hence $7 \mid n$, with witness $n - 2t$.

Compare with the failed cancellation modulo 6

Here the conclusion follows from a specific integer identity. Nonzero factors alone do not justify cancellation in congruences.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.5; numerical variant of Example 3.5.2.

The general cancellation statement behind the example

Theorem

Let $m, n, u, v, k \in \mathbb{Z}$, and suppose $um + vn = 1$. If $m \mid nk$, then $m \mid k$.

Proof. Choose $t \in \mathbb{Z}$ with $nk = mt$. Then

$$\begin{aligned} k &= (um + vn)k \\ &= m(uk) + v(nk) \\ &= m(uk) + v(mt) \\ &= m(uk + vt). \end{aligned}$$

The conclusion has witness $uk + vt$.

Source: Heather Macbeth, *The Mechanics of Proof*, Generalization of the witness method in Section 3.5.

Worked example: combine two divisibility assumptions

Suppose $5 \mid z$ and $7 \mid z$. Write $z = 5a$ and $z = 7b$. Since $3 \cdot 5 - 2 \cdot 7 = 1$,

$$\begin{aligned} z &= 15z - 14z \\ &= 15(7b) - 14(5a) \\ &= 35(3b - 2a). \end{aligned}$$

Thus $35 \mid z$.

Check the arithmetic relationship first

From $4 \mid 12$ and $6 \mid 12$, we cannot conclude $24 \mid 12$. The product conclusion needs more than two divisibility assumptions.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.5; numerical variant of Example 3.5.3.

Practice 3.5: find an identity equal to 1

1. Prove that $6 \mid 5n$ implies $6 \mid n$, for every integer n.
2. If $4 \mid z$ and $9 \mid z$, prove that $36 \mid z$.

Useful identities to investigate:

$$6 - 5 = 1, \qquad 9 - 2 \cdot 4 = 1.$$

State the new divisibility witness explicitly.

Practice 3.5: solution discussion

1. Choose $t \in \mathbb{Z}$ with $5n = 6t$. Then

$$n = 6n - 5n = 6n - 6t = 6(n - t).$$

The new witness is $n - t$.

2. Write $z = 4a$ and $z = 9b$. Then

$$z = 9z - 8z = 9(4a) - 8(9b) = 36(a - 2b).$$

The new witness is $a - 2b$.

Chapter 3: definitions turn number theory into algebra

Statement	What you must exhibit
n is even	$n = 2k$, with $k \in \mathbb{Z}$
n is odd	$n = 2k + 1$, with $k \in \mathbb{Z}$
$a \mid b$	$b = ak$, with $k \in \mathbb{Z}$
$a \equiv b \pmod{m}$	$a - b = mk$, with $k \in \mathbb{Z}$

- Use existing witnesses to build the witness required by the conclusion.
- Congruence respects addition, subtraction, multiplication, and natural powers.
- Cancellation needs a justified arithmetic condition; a Bézout identity supplies one.

Reading and source references

Heather Macbeth, *The Mechanics of Proof*, Chapter 3:

- 3.1 Definitions; parity
- 3.2 Divisibility
- 3.3 Modular arithmetic: theory
- 3.4 Modular arithmetic: calculations
- 3.5 Bézout's identity

https://hrmacbeth.github.io/math2001/03_Parity_and_Divisibility.html

Companion code: https://github.com/hrmacbeth/math2001

These slides independently present the definitions, proofs, and new numerical exercises. The power-rule induction is developed in Chapter 6.